



MATHEMATICS SSC-I
(Science Group)
SECTION – A (Marks 15)

Time allowed: 20 Minutes

Candidate Sign. _____

Invigilator Sign. _____

Answer Sheet No. _____

Version No.				
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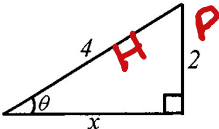
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Deleting/overwriting is not allowed.
Do not use lead pencil.

Fill the relevant bubble against each question according to curriculum:

Question	سوال	A	B	C	D	A	B	C	D
1. $\left(\frac{1}{16}\right)^{-\frac{3}{2}} =$		$\frac{1}{4}$	4	16	64	☐	☐	☐	☐
2. 4500 in scientific notation:		4.5×10^3	45×10^3	4.5×10^2	0.45×10^5	☐	☐	☐	☐
3. Which one is a binary relation?		{1,2,3}	{(1,2),(2,3)}	{1 → 2 → 3}	{1,2} × {3}	☐	☐	☐	☐
4. Highest common factor of $-12x^2y^2$, $6xy^3$, $24x^2y^2$ is:		$-6xy$	$-6x^2y^3$	$6xy^2$	$24y^3x^2$	☐	☐	☐	☐
5. $x+3 > 5$ implies:		$x=0$	$x < 2$	$x \leq 2$	$x > 2$	☐	☐	☐	☐
6.  $\theta = ?$		90°	60°	30°	0°	☐	☐	☐	☐
7. If a plane flies on a bearing of 180° , it is moving towards:		North	South	East	West	☐	☐	☐	☐
8. If two triangles are equal, then their corresponding sides are:		Equal	Unequal	Unproportional	Perpendicular	☐	☐	☐	☐
9. $\frac{1}{\sin \theta} =$		$\tan \theta$	$\cos \theta$	$\sec \theta$	$\operatorname{cosec} \theta$	☐	☐	☐	☐
10. The points where the medians of a triangle intersect is called the:		Incentre	Circumcentre	Centroid	Orthocentre	☐	☐	☐	☐
11. A triangular road sign has all sides equal. What type of triangle is it?		Right-angled	Equilateral	Isosceles	Scalene	☐	☐	☐	☐
12. The locus of points equidistant from two given points is the:		Perpendicular bisector	Angle bisector	Median	Altitude	☐	☐	☐	☐
13. Class mark of interval (15 – 20) is:		15	17.5	20	35	☐	☐	☐	☐
14. The probability of an event CANNOT be:		0	0.5	1	2	☐	☐	☐	☐
15. A boy scored 3 goals in 9 matches. His relative frequency of scoring goals is:		9	3	$\frac{1}{2}$	$\frac{1}{3}$	☐	☐	☐	☐

—1SA-I 26008 (B)—



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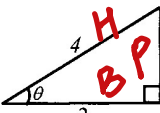
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Fill the relevant bubble against each question according to curriculum:

Question	A	B	C	D	A	B	C	D
1. $16^{\frac{1}{2}} =$	2	8	4	16	☐	☐	☐	☐
2. Correct scientific notation of 350 =	35×10^2	3.5×10^2	35×10^3	350×10^1	☐	☐	☐	☐
3. Domain of $\{(2,4), (3,5)\}$:	$\{2,3\}$	$\{4,5\}$	$\{3,4\}$	$\{2,4\}$	☐	☐	☐	☐
4. H.C.F of $8x^2$, $12x^3$ and $24x^2y^2$ is:	$4x^2$	$4x$	$8x^2$	$12x^3$	☐	☐	☐	☐
5. $x+5 > 9$ implies:	$x=0$	$x > 4$	$x < 4$	$x=4$	☐	☐	☐	☐
6.  $\theta =$	0°	30°	60°	90°	☐	☐	☐	☐
7. A ship moves from point A to B on a bearing of 90° . In which direction does it travel?	North	South	West	East	☐	☐	☐	☐
8. Two polygons are similar if they have:	Equal area	Equal perimeters	Equal corresponding angles	Equal volumes	☐	☐	☐	☐
9. Which of the following is a fundamental trigonometric identity?	$\sin \theta = \cos \theta$	$\sin^2 \theta + \cos^2 \theta = 1$	$\tan \theta = \sin \theta$	$\sec \theta = \cos \theta$	☐	☐	☐	☐
10. The angle bisectors of a triangle are concurrent at the:	Incentre	Orthocentre	Circumcentre	Centroid	☐	☐	☐	☐
11. The exterior angle of a regular polygon is 36° . number of sides in polygon are:	5	6	9	10	☐	☐	☐	☐
12. The locus of points at a fixed distance from a given point is a:	Straight line	Triangle	Circle	Altitude	☐	☐	☐	☐
13. The modal class in a grouped frequency distribution is the class with:	The highest frequency	The lowest frequency	The median frequency	The mean frequency	☐	☐	☐	☐
14. If an event has probability $\frac{3}{5}$, then the probability of the even not occurring is:	$\frac{2}{5}$	$\frac{3}{5}$	$\frac{5}{3}$	$\frac{1}{5}$	☐	☐	☐	☐
15. In a class, 18 out of 40 students received an A grade. The relative frequency of grade A is:	$\frac{6}{40}$	$\frac{18}{40}$	$\frac{2}{40}$	$\frac{8}{40}$	☐	☐	☐	☐

—1SA-I 26008 (D)—



MATHEMATICS SSC-I
(Science Group)

SECTION – A (Marks 15)

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Candidate Sign. _____

Supervisor Sign. _____

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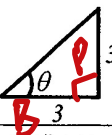
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Fill the relevant bubble against each question according to curriculum:

Question	سوال	A	B	C	D	A	B	C	D
1. $\left(\frac{1}{9}\right)^{-\frac{1}{2}}$ equals:		3	6	9	81	☐	☐	☐	☐
2. Scientific notation of 71000 is:		7.1×10^2	0.71×10^2	7.1×10^3	7.1×10^4	☐	☐	☐	☐
3. Range of $\{(1,3), (2,5)\}$ is:		$\{1,2\}$	$\{3,5\}$	$\{1,3\}$	$\{2,5\}$	☐	☐	☐	☐
4. For two polynomials A and B, $HCF \times LCM =$		$A+B$	$A-B$	$(HCF)^2$	$A \times B$	☐	☐	☐	☐
5. $-2x \leq 4$ gives:		$x \leq -2$	$x \geq -2$	$x \leq -1$	$x = -1$	☐	☐	☐	☐
6.  $\theta = ?$		45°	60°	90°	120°	☐	☐	☐	☐
7. A cyclist travels from P to Q on a bearing of 45° . This means the cyclist moves.		South-East	North-East	South-West	North-West	☐	☐	☐	☐
8. If two triangles are equal, then their corresponding sides are:		Equal	Unequal	Unproportional	Perpendicular	☐	☐	☐	☐
9. $1 + \tan^2 \theta =$		$\sin^2 \theta$	$\cos^2 \theta$	$\sec^2 \theta$	$\operatorname{cosec}^2 \theta$	☐	☐	☐	☐
10. The perpendicular bisectors of the sides of a triangle meet at the:		Incentre	Centroid	Orthocentre	Circumcentre	☐	☐	☐	☐
11. Each interior angle of a regular square-shaped garden is:		90°	60°	45°	30°	☐	☐	☐	☐
12. The locus of points equidistant from two intersecting straight lines is formed by the:		Two parallel lines	Perpendicular bisectors	Circles	Angle bisectors	☐	☐	☐	☐
13. The median of 17, 27, 37, 47, 57, 67, 77 is:		47	17	77	0	☐	☐	☐	☐
14. If a fair die is rolled once, what is the probability of getting 2?		$\frac{1}{2}$	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{5}$	☐	☐	☐	☐
15. Which is a possible Valid relative frequency value:		-0.5	1.2	0.75	2	☐	☐	☐	☐

—ISA-I 26008 (M)—



MATHEMATICS SSC-I (Science Group)

SECTION – A (Marks 15)

Time allowed: 20 Minutes

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Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر نام مرکز کے حوالے کریں۔ گات کر دیا ہو
کھینچنے کی اجازت نہیں ہے۔ لپیڈ پینسل کا استعمال ممنوع ہے۔

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ROLL NUMBER							

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Answer Sheet No. _____

Invigilator Sign. _____

Fill the relevant bubble against each question according to curriculum:

Candidate Sign. _____

Question	سوال	A	B	C	D	A	B	C	D
1. Which of the following is equal to $(x^{\frac{2}{3}})^9$?	Which of the following is equal to $(x^{\frac{2}{3}})^9$?	x^8	x^6	$x^{\frac{4}{9}}$	$x^{\frac{3}{2}}$	☐	☐	☐	☐
2. 5.67×10^4 in standard form is:	5.67×10^4 کو معیاری صورت میں ظاہر کریں:	0.000567	0.567	567000	56700	☐	☐	☐	☐
3. What is the domain of the relation $R = \{(1,2), (2,3), (3,4)\}$?	What is the domain of the relation $R = \{(1,2), (2,3), (3,4)\}$?	$\{1,2,3\}$	$\{2,3,4\}$	$\{1,3,4\}$	$\{2,4\}$	☐	☐	☐	☐
4. HCF of $3x^2$ and $6x^3$ is:	HCF of $3x^2$ and $6x^3$ is:	$3x$	$6x^2$	$3x^2$	$3x^3$	☐	☐	☐	☐
5. The solution to the inequality $x + 3 < 5$ is:	The solution to the inequality $x + 3 < 5$ is:	$x > 2$	$x < 2$	$x = 2$	$x \geq 2$	☐	☐	☐	☐
6. Conversion of 60° in radians is:	Conversion of 60° in radians is:	$\frac{\pi}{6}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	☐	☐	☐	☐
7. If the bearing of B from A is 135° , what direction is it?	If the bearing of B from A is 135° , what direction is it?	NW	SE	SW	NE	☐	☐	☐	☐
8. Two triangles having equal corresponding angles are called:	Two triangles having equal corresponding angles are called:	Similar triangles	Right triangles	Scalene triangles	Isosceles triangles	☐	☐	☐	☐
9. Which of the following identities is correct?	Which of the following identities is correct?	$\sin^2 \theta + \cos^2 \theta = 2$	$\sin^2 \theta + \cos^2 \theta = 1$	$\tan^2 \theta = 1 + \sec^2 \theta$	$\cos^2 \theta - \sin^2 \theta = 0$	☐	☐	☐	☐

A O

Question	سوال	A	B	C	D	A	B	C	D
10. The point where all altitudes of a triangle meet is called:	وہ نقطہ جہاں مثلث کے تمام عمود ملتے ہیں، کہا جاتا ہے۔	Incentre ان سنٹر	Centroid سنٹر انڈ	Orthocenter آرٹھو سنٹر	Circumcenter سرکمر سنٹر	☐	☐	☐	☐
11. Each interior angle of a regular octagon is equal to:	ایک باقاعدہ ہشت پہلو شکل کا ہر اندرونی زاویہ _____ ہوتا ہے۔	108°	120°	135°	140°	☐	☒	☐	☐
12. The locus of points equidistant from two given points is:	دو دیے گئے نقاط سے برابر فاصلے پر تمام نقاط کا locus _____ کہا جاتا ہے۔	Angle bisector ناصف زاویہ	Line joining the points دونوں نقاط کو جوڑنے والی کثیر	Perpendicular bisector عمود کی ناصف	Median میڈین	☐	☐	☐	☐
13. Median of 5, 7, 8, 10, 12 is:	5, 7, 8, 10, 12 کا وسطانیہ (Median) کیا ہے؟	8	9	10	7	☐	☐	☐	☐
14. Probability of drawing a heart from a deck of 52 cards is:	52 کارڈز کے سیٹ سے ہارٹ نکالنے کا امکان کیا ہے؟	$\frac{1}{4}$	$\frac{1}{13}$	$\frac{1}{8}$	$\frac{1}{2}$	☒	☐	☐	☐
15. Relative frequency of 8 successes out of 40 trials is:	40 کوششوں میں 8 کامیابیوں کی ریلیٹو فریکوئنسی کیا ہے؟	0.1	0.2	0.4	0.8	☐	☐	☐	☐

—1SA-II 25008—

ROLL NUMBER



MATHEMATICS SSC-I (Science Group)

SECTION – A (Marks 15)

Time allowed: 20 Minutes

Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed.

Do not use lead pencil.

حصہ اول لازمی ہے اس کے جوابات اسی صفحہ پر دے کر تمام مرکز کے حوالے کریں۔ کات کر دوبارہ لکھنے کی اجازت نہیں ہے۔ سیاہ قلم کا استعمال نہ کریں۔

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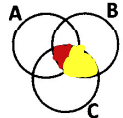
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Answer Sheet No. _____

Invigilator Sign. _____

Fill the relevant bubble against each question according to curriculum:

Candidate Sign. _____

Question	سوال	A	B	C	D	A	B	C	D
1. $\left(\frac{1}{4}\right)^{-\frac{1}{2}} =$		$\frac{1}{2}$	2	$\sqrt{2}$	16	☐	☐	☐	☐
2. The scientific notation of 345.6 is: 345.6 کی سائنٹفک نوٹیشن کیا ہے گی؟		3.456×10^2	3.456×10^3	3.456×10^{-2}	3.456×10^{-3}	☐	☐	☐	☐
3. Given relation R is a subset of _____, if: $X=\{2,4,6\}$ $Y=\{1,3,5\}$ $R=\{(1,2),(3,4),(5,6)\}$ دیا گیا ریلیشن R کس کا سب سیٹ ہو گا؟		$X \times X$	$X \times Y$	$Y \times X$	$Y \times Y$	☐	☐	☐	☐
4. What statement does the shaded region represent? شیدڈ ریجن کس سٹیٹمنٹ کو ظاہر کرتا ہے؟		$A \cap B \cap C$	$(A \cap B) \cap C$	$B \cap C$	$A \cup C$	☐	☐	☐	☐
5. Factorization of: $49a^2-28ab+4b^2=$ دیے گئے فقرے کی تجزیہ:		$(7a+2b)^2$	$(49a-4b)^2$	$(7a-2b)^2$	$(7a-28b)^2$	☐	☐	☐	☐
5. If x is smaller than 10, then which statement is TRUE? اگر x عدد 10 سے چھوٹا ہے تو کون سی سٹیٹمنٹ درست ہے؟		$x \geq 20$	$x \leq 10$	$x < 10$	$x > 10$	☐	☐	☐	☐
7. Slope of a line passing through two points (3,5) and (9,11) is: نقطہ (3,5) اور (9,11) سے گزرتی لائن کی سلوپ:		$\frac{8}{3}$	$\frac{4}{3}$	$\frac{6}{6} = 1$	$\frac{8}{4}$	☐	☐	☐	☐
3. At what angle do the given lines intersect each other? $y = \frac{2}{3}x - 4$ $y = \frac{-3}{2}x + 4$ دی گئی لائنز ایک دوسرے کو کس زاویے پر قطع کرتی ہیں؟		0°	45°	90°	180°	☐	☐	☐	☐



Question	A	B	C	D	A	B	C	D
9. What is the value of x in the given similar triangles? دی گئی متشابه مثلثوں میں x کی ویلیو کیا ہے؟	1	2	3	6	☐	☐	☐	☐
10. What is the value of measure angle C in the figure? دی گئی شکل میں زاویہ C کی مقدار ارکتی ہے؟	29.1°	60.94°	56.25°	33.75°	☐	☐	☐	☐
11. The bearing of A from O is: نقطہ A کی بیرنگ کیا ہے؟	035°	055°	125°	305°	☐	☐	☐	☐
12. The three medians of any triangle are: کسی مثلث کے وسطانیے _____ ہوتے ہیں۔	Concurrent inside	Concurrent outside	Collinear	Mutually perpendicular	☐	☐	☐	☐
13. The median of the given data is: 77, 79, 82, 86, 90, 92, 93 دیے گئے مواد کا وسطانیہ کیا ہوگا؟	79	82	86	88	☐	☐	☐	☐
14. What is the probability of getting Tail, if a fair coin is tossed once? اگر ایک مناسب سکہ اچھالا جائے تو 'ٹیل' آنے کا کتنا امکان ہے؟	0	$\frac{1}{2}$	1	2	☐	☐	☐	☐
15. A die was rolled 50 times, and the number 4 appeared 15 times. What is the relative frequency of rolling a 4? لڈو کا ایک دانہ 50 مرتبہ پھینکنے پر 15 مرتبہ نمبر 4 آیا۔ نمبر 4 کی ریلیٹیو فریکوئنسی کیا ہوگی؟	0.15	0.30	0.20	0.25	☐	☐	☐	☐

15/50

—1SA-I 25008-10081(B)—

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MATHEMATICS SSC-I (Science Group)

SECTION – A (Marks 15)

Time allowed: 20 Minutes

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حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر نام مرکز کے حوالے کریند کات کر دو ہمارے
لکھنے کی اجازت نہیں ہے۔ اس پر غلط استعمال ممنوع ہے۔

Version No.				
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Answer Sheet No. _____

ہر سوال کے سامنے دیے گئے، کریکولم کے مطابق درست دائرہ کو پر کریں۔

Invigilator Sign. _____

Fill the relevant bubble against each question according to curriculum:

Candidate Sign. _____

Question	سوال	A	B	C	D	A	B	C	D
1.	$\left(\frac{1}{9}\right)^{-\frac{3}{2}} =$	27	$\frac{1}{27}$	± 27	$\pm \frac{1}{27}$	☐	☐	☐	☐
2.	The scientific notation of 0.00567 is:	5.67×10^3	5.67×10^{-3}	5.67×10^{-2}	5.67×10^2	☐	☐	☐	☐
3.	For sets $A = \{a, b\}$ and $B = \{1, 2\}$ then $R = \{(a, 1), (b, 2)\}$ is a subset of:	$A \times A$	$A \times B$	$B \times A$	$B \times B$	☐	☐	☐	☐
4.	In the Venn diagram, region $A \cap B \cap C$ represents elements in:	A and B but not in C	A, B and C	A or B but not C	A only	☐	☐	☐	☐
5.	Factorization of: $4a^2 - 12ab + 9b^2 =$	$(2a-3b)^2$	$(2a+3b)^2$	$(4a+3b)^2$	$(4a-3b)^2$	☐	☐	☐	☐
6.	A pizza costs Rs. 1350, and you have Rs. 2500. Maximum how many/much pizzas can be bought with this money?	≥ 1.85	≤ 1.85	> 1.85	2	☐	☐	☐	☐
7.	Slope of a line passing through two points (4, 0) and (0, -8) is:	-2	0.5	2	0	☐	☐	☐	☐
8.	The angle between the given lines is: $y = \frac{3}{4}x - 5$ $y = \frac{3}{4}x + 5$ is:	0°	45°	90°	180°	☐	☐	☐	☐
9.	If two polygons are similar in shape and ratio of their corresponding sides is 2:3, what is the ratio of their areas?	2:3	4:9	8:27	1:1	☐	☐	☐	☐

$$\sin 35^\circ = \frac{5}{x} \Rightarrow x = \frac{5}{\sin 35^\circ} \approx 7.7$$

Question	A	B	C	D	A	B	C	D
10. What is the value of x in the given figure? دی گئی شکل میں x کی مقدار کتنی ہے؟	5cm	10cm	7.5cm	15cm	☐	☐	☐	☐
11. A ship sails on a three-figure bearing of 075° . In which direction is the ship heading? ایک کشتی بحری گریڈنگ پر 075° چل رہی ہے۔ اس کا رخ کیا ہوگا؟	North شمال	East مشرق	North East شمال مشرق	South East جنوب مشرق	☐	☐	☐	☐
12. In how many equal-area triangles, do the three medians divide a triangle? دسٹائیے کسی مثلث کو کتنی مساوی رقبہ کی چھوٹی مثلثوں میں تقسیم کرتے ہیں؟	3	4	5	6	☐	☐	☐	☐
13. A company surveyed the number of products purchased in a month: {2, 3, 3, 2, 4, 5, 3, 6}. What is the mode of the data? ایک کمپنی نے پراڈکٹس کی خرید کا سروے کیا۔ جس میں درج شدہ سامان آئے۔ {2, 3, 3, 2, 4, 5, 3, 6} اس مواد کا ماہرہ معلوم کریں۔	3	4	5	6	☐	☐	☐	☐
14. A bag contains 4 blue balls and 6 red balls. If one ball is drawn at random, what is the probability that it is NOT blue? ایک بیگ میں 4 نیلی اور 6 سرخ گیندیں ہیں۔ اگر ایک گیند نکالی جائے تو اس بات کا کتنا امکان ہے کہ نکالی جانے والی گیند نیلی نہیں ہے؟	$\frac{2}{5}$	$\frac{3}{5}$	$\frac{4}{5}$	$\frac{6}{5}$	☐	☐	☐	☐
15. In 50 trials, an event occurs 10 times. What is the relative frequency of the event? 50 کوششوں میں ایک واقعہ 10 مرتبہ وقوع پذیر ہوتا ہے۔ اس واقعہ کی ریلیٹیو فریکوئنسی کتنی ہے؟	$\frac{1}{5}$	$\frac{1}{10}$	$\frac{1}{2}$	$\frac{1}{50}$	☐	☐	☐	☐

—1SA-I 25008-12081(D)—

ROLL NUMBER					



Federal Board SSC-I Examination Model Question Paper Mathematics

(Curriculum 2022-23)

Section - A (Marks 15)

Time Allowed: 20 minutes

Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent.

Deleting/overwriting is not allowed. Do not use lead pencil.

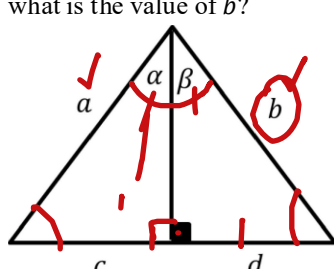
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Candidate Sign. _____

Invigilator Sign. _____

Q1. Fill the relevant bubble against each question. Each part carries one mark.

Sr no.	Question	A	B	C	D	A	B	C	D
i.	The radical form of $x^{-\frac{3}{2}}$ is:	$\sqrt[3]{x^2}$	$\frac{1}{\sqrt{x^3}}$	$\sqrt{x^3}$	$\frac{1}{\sqrt[3]{x^2}}$	☐	☐	☐	☐
ii.	Which of the given options represents the scientific notation of 0.25^2 ?	625×10^{-4}	62.5×10^{-3}	6.25×10^{-2}	0.625×10^{-1}	☐	☐	☐	☐
iii.	If $A = \{2,4,6\}$ and $B = \{0,1\}$, then find number of elements in $A \times B$.	5	6	8	9	☐	☐	☐	☐
iv.	What is the least common multiple of $7x - 6xy$ and $5xy^3 - 3x^2$?	$(7 - 6y) \times (5x^3 - 3x)$	$(7x - 6xy) \times (5y^3x - 3x^2)$	$x(7 - 6y)$	$x(7 - 6y) \times (5y^3 - 3x)$	☐	☐	☐	☐
v.	Solution of inequality $-2x - \frac{1}{2} \leq \frac{3}{2}$ is:	$x > -1$	$x < -1$	$x \geq -1$	$x \leq -1$	☐	☐	☐	☐
vi.	What is the radian measure of $15^\circ 50'$?	$\frac{19\pi}{216}$	$\frac{19\pi}{36}$	$\frac{19\pi}{180}$	$\frac{216\pi}{19}$	☐	☐	☐	☐
vii.	If a navigator gives bearing 0° , in which direction should he travel?	North	South	East	West	☐	☐	☐	☐
viii.	In the figure, if $\alpha = \beta$ then what is the value of b ? 	$\frac{cd}{a}$	$\frac{c}{ad}$	$\frac{ad}{c}$	$\frac{ac}{d}$	☐	☐	☐	☐

ix.	What is the value of $-3 - 3 \tan^2 \theta$ in a single trigonometric function?	$3 \operatorname{cosec}^2 \theta$	$-3 \sec^2 \theta$	$3 \sec^2 \theta$	$-3 \operatorname{cosec}^2 \theta$	☐ ☐ ☐ ☐
x.	Which of the following points is the intersection of the angle bisectors of a triangle?	circumcenter	orthocenter	incentre	centroid	☐ ☐ ☐ ☐
xi.	Each of the internal angle of a regular hexagon is:	60°	72°	108°	120°	☐ ☒ ☐ ☐
xii.	Locus of points equidistant from $P(5,4)$ and $Q(5,-6)$ is:	$x = 0$	$x = 5$	$y = -1$	$y = 1$	☐ ☐ ☐ ☐
xiii.	The mean of 11 numbers is 7. One of the numbers 13 is deleted. What is the mean of the remaining 10 numbers?	7.7	6.4	6.0	5.8	☐ ☐ ☐ ☐
xiv.	What is the probability of picking a king from well-shuffled 52 playing cards?	$\frac{1}{52}$	$\frac{1}{13}$	$\frac{4}{13}$	$\frac{1}{26}$	☐ ☐ ☐ ☐
xv.	A fair coin is tossed twice, then the frequency of appearing head twice is:	$\frac{1}{4}$	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{3}{4}$	☐ ☐ ☐ ☐

(HH) HT, TH, TT

$\frac{4}{52}$

$\frac{1}{4}$



Federal Board SSC-I Examination Mathematics Model Question Paper

National Curriculum 2022-2023 (Inclusive Scheme of Studies 2024)
(50% MCQs & 50% Descriptive)

Section - A (Marks 36)

Time Allowed: 1:20 hours

Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent.

Deleting/overwriting is not allowed. Do not use lead pencil.

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Version No.			
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Invigilator Sign. _____

Candidate Sign. _____

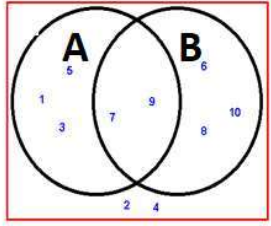
Note: Fill the correct bubble for each question. Each question carries 0.75 mark.

Sr no.	Question	A	B	C	D	A	B	C	D
1	The radical form of $2^3 \times 27$ is:	$3 \times \sqrt[3]{2^4}$	$3^3 \times \sqrt[4]{2^3}$	$3^3 \times \sqrt[3]{2^4}$	$3^4 \times \sqrt[4]{2^4}$	☐	☐	☐	☐
2	The scientific notation of 150 million kilometres is:	1.5×10^6	1.5×10^8	1.5×10^{-6}	1.5×10^{-8}	☐	☐	☐	☐
3	The temperature in a freezer was -9°C . It increased by $\frac{5^\circ}{2}\text{C}$. What is the new temperature?	7°C	2°C	-2°C	-7°C	☐	☐	☐	☐
4	For what values of x and y , $(x + y, 4x + 1) = (3, 9)$?	$x = 1, y = 2$	$x = 0.5, y = 8.5$	$x = 2, y = 1$	$x = 2, y = 7$	☐	☐	☐	☐
5	If $(x - 1)$ is a factor of $x^3 - 5x^2 + ax + 6$, the value of a is:	-2	-1	0	2	☐	☐	☐	☐
6	If $x^2 - 11x + 30 = (x - a)(x - b)$, then value of $a + b$ is:	-30	-11	11	30	☐	☐	☐	☐
7	The HCF of $6x^2y$ and $9xy^2$ is:	$3x^2y$	$3x^2y^2$	$3xy$	$18x^2y^2$	☐	☐	☐	☐
8	If $\sqrt{x^2 + 6x + 9} = x + k$, the value of k is:	-3	$\pm\sqrt{9}$	3	$\pm\sqrt{3}$	☐	☐	☐	☐
9	The profit of a small business is modelled by $x^2 - 9x + 20$. For a breakeven point the factorized expression is:	$(x - 4)(x - 5)$	$(x - 4)(x + 5)$	$(x + 4)(x - 5)$	$(x + 4)(x + 5)$	☐	☐	☐	☐
10	On solving $\frac{x}{2} - \frac{3}{2} \geq 2$, the value of x is:	$x \leq 5$	$x \geq 5$	$x \geq 7$	$x \leq 7$	☐	☐	☐	☐
11	Which value of y satisfies the equation $0.6y + 1.2 = 3$?	0.3	1.08	3	7	☐	☐	☐	☐
12	What is the midpoint of the line segment joining A (2, 4) and B (6, 10)?	(2,3)	(4,7)	(6,5)	(8,14)	☐	☐	☐	☐

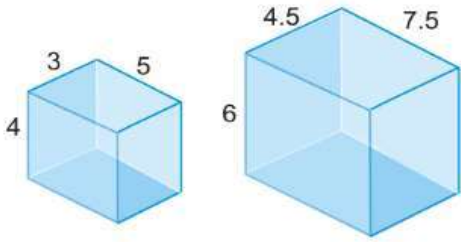
13	What is the gradient of a line passing through points (3, 7) and (3, -5)?	0	-2	$\frac{1}{3}$	undefined	☐	☐	☐	☐
14	If $y = 2x + c$ passes through the point (3, 10), what is the numerical value of c ?	-17	4	16	23	☐	☐	☐	☐
15	The line $y = 5x + 1$ is perpendicular to another line. What is the gradient of the other line?	5	-5	$\frac{1}{5}$	$-\frac{1}{5}$	☐	☐	☐	☐
16	Two schools are located at points (2, 3) and (8, 11) on a map. What is the distance between them?	100 units	14 units	10 units	5 units	☐	☐	☐	☐
17	On a toy model car, the length of the door is 3 cm. If the actual car is exactly 40 times larger than the model, what is the length of the door on the actual car?	43 cm	80 cm	120 cm	1200 cm	☐	☐	☐	☐
18	A bakery makes two sizes of similar cube-shaped cakes. The larger cake has sides that are twice as long as the smaller cake. If the small cake requires 1 cup of butter, how many cups of butter are needed for the larger cake?	2 cups	4 cups	8 cups	16 cups	☐	☐	☐	☐
19	Two intersecting roads form supplementary angles where one angle is three times the other. What is the measure of the smaller angle?	22.5°	45°	90°	135°	☐	☐	☐	☐
20	Two sides of a triangle have slopes 1 and -1. What is the angle between them?	45°	90°	135°	180°	☐	☐	☐	☐
21	Two similar triangles have corresponding sides in the ratio 1 : 3. What is the ratio of their areas?	1 : 6	6 : 1	9 : 1	1 : 9	☐	☐	☐	☐
22	A cuboid has volume 1000 cm^3 . It is enlarged by a scale factor of 2. What is the volume of the enlarged cuboid?	500 cm^3	2000 cm^3	4000 cm^3	8000 cm^3	☐	☐	☐	☐
23	A goat is tied to a corner of a square field with a 7m rope. The locus of points it can graze is a:	Quarter circle	Semi-circle	Full circle	Square	☐	☐	☐	☐
24	A dog is tied to a fixed pole in a park with a 5 m rope. The set of all points the dog can reach is a:	Circle	Semi-circle	Square	Rectangle	☐	☐	☐	☐
25	135° in circular system (radians) is:	$\frac{3\pi}{4}$	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{4}$	☐	☐	☐	☐
26	A person stands 20 m from a building. The angle of elevation to the top is 45° . What is the height of the building?	11.5 m	14.1 m	20 m	34.6 m	☐	☐	☐	☐
27	The value of $\cos^2 100^\circ + \sin^2 100^\circ$ is:	0	0.81	1	1.15	☐	☐	☐	☐

28	A ship sails from port on a bearing of 225° . In which direction is it moving?	North-East	South-East	South-West	North-West	☐	☐	☐	☐
29	In a triangle, the three medians meet at a point called the:	Circumcenter	Orthocenter	Centroid	Incentre	☐	☐	☐	☐
30	The vertices of a right-angled triangle are (0, 0), (6, 0), and (0, 8). What is the midpoint of the hypotenuse?	(3, 4)	(6, 8)	(3, 0)	(0, 4)	☐	☐	☐	☐
31	What is the probability of getting M in MATHEMATICS?	$\frac{1}{10}$	$\frac{2}{10}$	$\frac{1}{11}$	$\frac{2}{11}$	☐	☐	☐	☐
32	A fair coin is tossed 100 times. The expected number of Tails to occur is:	0	1	50	100	☐	☐	☐	☐

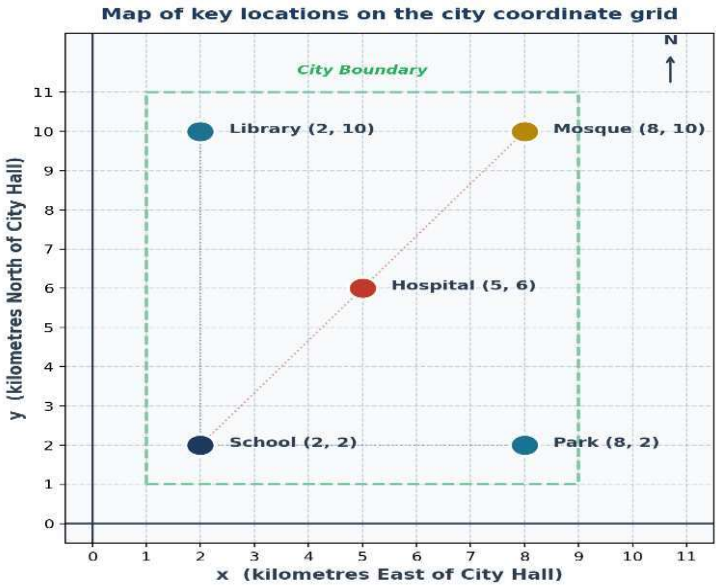
STIMULUS/SITUATION/SCENARIO BASE MCQs

	Scenario for MCQs 33 to 36	The Venn diagram shows the elements of set A and set 							
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33	Which of the elements belong only to set A?	{1,2,4,6, 7}	{1,3,5,7,9}	{1,3,5}	{7,9}	☐	☐	☐	☐
34	Which of the elements are in the intersection of sets A and B?	{1,3,5, ...,10}	{7,9}	{1,3,5}	{6,8,10}	☐	☐	☐	☐
35	Which of the elements are not included in either set A or set B?	{1,2,3, ...,10}	{1,3,5}	{2,4}	{6,8,10}	☐	☐	☐	☐
36	The elements of $A - B$ are:	{1,3,5}	{6,8,10}	{7,9}	{2,4}	☐	☐	☐	☐

	Scenario for MCQs 37 to 40	The given prisms are similar 							
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37	What is the linear scale factor from the smaller cuboid to the larger cuboid?	1.25	1.5	2.0	2.25	☐	☐	☐	☐
38	The total surface area of the smaller cuboid is 94 cm^2 , what is the total surface area of the larger cuboid?	141 cm^2	211.5 cm^2	317.25 cm^2	423 cm^2	☐	☐	☐	☐
39	The volume of the smaller cuboid is 60 cm^3 , what is the volume of the larger cuboid?	90 cm^3	135 cm^3	202.5 cm^3	270 cm^3	☐	☐	☐	☐
40	What is the ratio of the volume of the smaller prism to the volume of the larger prism?	1 : 1.5	1 : 2.25	1 : 3.375	1 : 4.5	☐	☐	☐	☐

	Scenario for MCQs (41-44)	The table below shows weight distribution (in kg) of students in a class.							
		C-I	30-40	40-50	50-60	60-70	70-80		
		f	6	10	16	12	6		
41	What is the size of class interval?	10	11	12	15	☐	☐	☐	☐
42	What is the value of $\frac{\Sigma f}{2}$?	20	22	25	26	☐	☐	☐	☐
43	What is the cumulative frequency for the class 50 – 60?	16	26	32	34	☐	☐	☐	☐
44	Which is the median class?	40 – 50	50 – 60	60 – 70	30 – 40	☐	☐	☐	☐
Scenario for MCQs (45-48)		Study the given map and answer the following questions. 							
45	What is the distance between School and Park?	4 km	6 km	8 km	10 km	☐	☐	☐	☐
46	What is the midpoint of the line segment joining Library and Park?	(4, 5)	(5, 5)	(5, 6)	(6, 5)	☐	☐	☐	☐
47	What is the slope of the line joining two points (school and mosque)?	$\frac{4}{3}$	$\frac{3}{4}$	$\frac{6}{5}$	$\frac{5}{6}$	☐	☐	☐	☐
48	Which pair of locations on the map is exactly 8 km apart?	School & Hospital	Park & Hospital	Park & Mosque	Library & Hospital	☐	☐	☐	☐